Sentinel-52: a receiving-desk decision tool for a regional food bank

NAFSI Hub Student Data Challenge, Fall 2026, SENS-D track. Rushikesh Lagad (University of Arkansas), Sambhavi Patel (University of Arkansas), Naisargi Modha (Northeastern University).

Live application: https://rushilagad.github.io/sentinel-52/

Repository: https://github.com/RushiLagad/sentinel-52

How it was built, in brief

Everything in this report is produced by `analysis/run_all.py` from the raw FSIS files and the public endpoints named in `data/PROVENANCE.md`, and every number quoted below names the output file it comes from. The pipeline has four parts.

Data. Thirteen fiscal-year archives of FSIS establishment-specific raw poultry sampling (325,123 samples, FY2014 to FY2026), the FSIS performance-standard category postings, NARMS cecal sampling, the inspection directory and the establishment demographic file, and CDC NORS from its public endpoint. Added to those: NOAA monthly climate, the FSIS recall API, CDC/ATSDR SVI 2022, and a seeded synthetic lot inventory. Raw files are verified against SHA-256 hashes before anything is built.

Model. The unit is a plant and product class on FSIS's own 52-week window; the outcome is entry into Category 3 within a horizon of 3, 6 or 12 months. The category rebuild was validated against three official postings first. Three scorers are compared, a transparent binomial projection, logistic regression and gradient boosting, on a time-ordered split (train through September 2022, test from January 2024). Validation reports AUROC and average precision with establishment-clustered bootstrap intervals, reliability and expected calibration error per scorer, lead time, watch-list comparison at equal list size, and a leave-one-group-out ablation.

Decision layer. Each simulated lot is matched to the most recent score available on its receiving date (never a later one, and none older than 120 days), then assigned one of four actions under a policy: distribute now, prioritize, hold for review, or divert to the cooked-meal channel. Four policies are costed in the same units, meals at 1.2 lb each, with the two loss mechanisms derived from shelf life and cooked-channel capacity rather than assumed. The scorer is picked per horizon by measured calibration.

Application. A React app reads one JSON file built from those outputs, with four guardrails enforced at build time and in 17 tests: no safe/unsafe language, no establishment beside a burden figure, every simulated quantity marked, and every high-consequence action behind an acknowledgement that shows the evidence that would reverse it.

Reproduction instructions, including the manual download that the FSIS content network now requires, are in the README. The decision layer reproduced to the digit on two machines.

Q1. The decision-maker and the decision horizon

The brief asks for one supply-chain actor and a decision that actor actually makes. We chose a regional food bank in Arkansas, and the choice was about the challenge question rather than convenience. The question asks how an actor can protect health while preserving food availability, supporting food security, and minimizing avoidable food loss. For a processor or a retailer those three are constraints on a safety decision. For a food bank they are the mission, and they collide with protecting health every day. Discard a pallet of chicken on a weak signal and people with no alternative go without; distribute it on a missed signal and the people harmed are the least able to absorb harm. For this actor the conflict the brief asks us to surface is the daily job, not a hypothetical.

The decision is taken per lot, inside the lot's perishability window: for each incoming or on-hand lot of FSIS-regulated poultry, distribute now, prioritize for immediate distribution, hold for review, or divert. The mechanism that makes this possible is ordinary. The USDA mark of inspection on every case carries the establishment number, so a pallet traces to the plant that produced it, and that plant has a public sampling history going back to fiscal year 2014.

Divert does not mean discard. A plant with a deteriorating sampling trajectory does not make its product inedible; it makes uncooked distribution the wrong channel for it. Diverting to a partner that cooks on site keeps the food in the system and applies a validated kill step. That is the actor-specific move that lets the health objective and the loss objective be traded against each other rather than simply opposed. It is also the reason the tool is built for a receiving desk and not as a recall simulator.

The forecast horizon is three months. Two reasons, one about the decision and one we did not anticipate. The decision is taken within days, so the nearest horizon is the relevant one. And coverage: 80 percent of lots have a usable score at the three-month horizon against 64 percent at six months, because the six-month test window ends earlier and more lots fall past the staleness limit. An unscored lot is distributed with the gap named, so coverage feeds directly into recall. Run at six months the conservative policy reaches recall 0.272 and destroys 62,042 meals; at three months it reaches 0.403 and destroys 30,529. Three months is better on both axes at once (`report/FINDINGS.md`, Finding 4).

The tool ranks what to look at first and puts a cost on each choice. It is not a regulatory determination, not an outbreak declaration, and not a statement that any food, lot or plant is safe or unsafe.

Q2. The role of each core dataset

Five FSIS datasets and one CDC dataset carry the project. Each is documented in `data/PROVENANCE.md` with its source, access date, licence, coverage, the transformations we applied and the limitations we know about, and every column of every table we built from them is in `data/DATA_DICTIONARY.md`. Here is what each one does.

FSIS establishment-specific raw poultry sampling, FY2014 to FY2026. The backbone. 325,123 laboratory samples with 37,685 Salmonella positives across 1,409 establishments that were ever sampled, each sample carrying the plant, the product class, the date, the

result, and for positives the serotype, the whole-genome resistance calls and, from 2022, the enumeration. Every risk score, every 52-week count and every outcome label in the project comes from this file. Thirteen fiscal-year archives, 62 MB, verified against SHA-256 hashes in `data/MANIFEST.json` before anything is built.

FSIS Salmonella performance-standard categories. The posted Category 1, 2, 3 and NA ratings. We use them twice. They are the status-quo signal a food bank would act on today, and so the rule-based baseline. And they are the validation target: before modelling anything we rebuilt the categories from the raw samples and checked the rebuild against three official postings, which agree on 99.75 percent of 2,764 rows with zero false Category 3s. A tool that could not reproduce FSIS's own rating would have no business forecasting it.

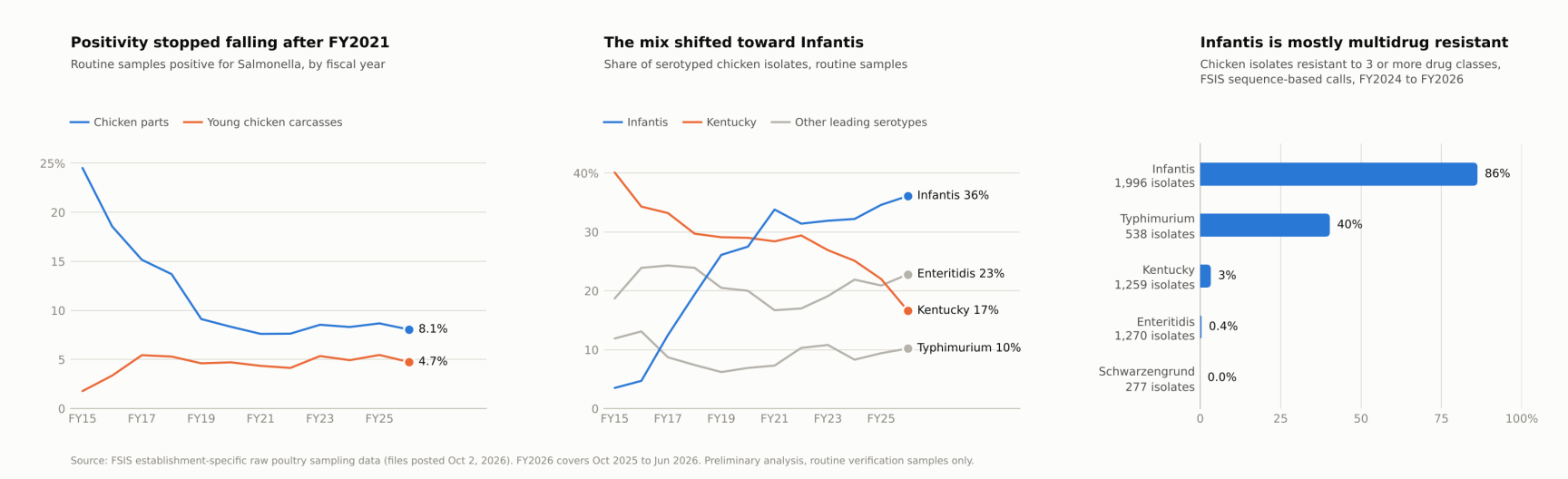


Figure 1: Routine positivity by fiscal year and product class; the shift of the chicken serotype mix toward Infantis; and the share of Infantis isolates resistant to three or more drug classes by FSIS sequence-based calls. Built by 07_figures.py from t1, t2b and t13.

FSIS NARMS cecal sampling. 72,151 cecal samples at slaughter with resistance detail. This is the animal-health context for the serotype story in Finding 1, and what lets us say that the serotype FSIS finds most often is not the one that sends people to hospital.

The Meat, Poultry and Egg Product Inspection Directory and the establishment demographic file. 8,188 establishments, active and inactive, with location, HACCP size, slaughter and processing flags, and the county FIPS code. This is the join from the number on a pallet's mark of inspection to a real plant, and from the plant to a county.

CDC NORS, from the public Socrata endpoint. The human-illness layer. Outbreaks, illnesses, hospitalisations and deaths by IFSAC food category and serotype. The brief permits a public CDC human-illness dataset in place of PulseNet, and NORS is one. It enters the project at serotype and year only. We did not use the NORS file the organisers circulated, because it was restricted to the team and the public endpoint carries the same attribution field.

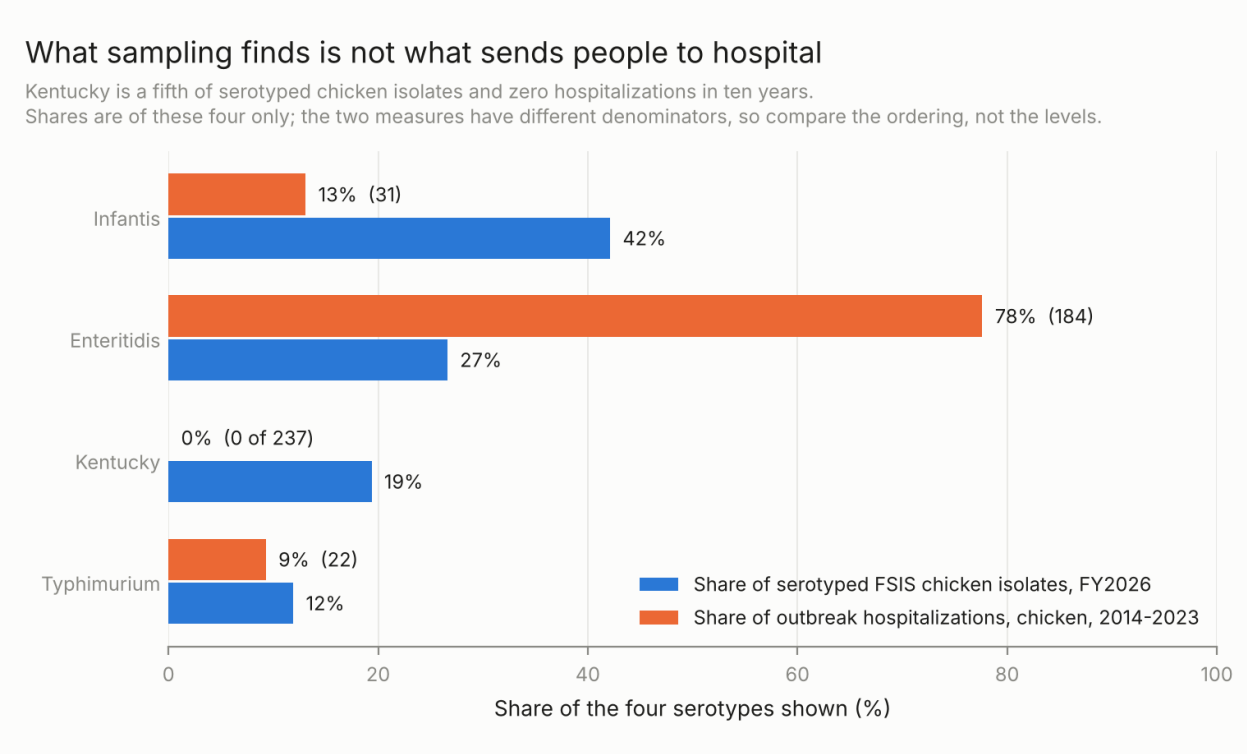


Figure 2: Serotype share of FSIS chicken isolates against share of NORS poultry outbreak hospitalisations, the four serotypes normalised so the ordering is comparable. Joined at serotype and year only. Built by 10b_figure_prevalence_vs_burden.py from t17.

The licence on the FSIS data forbids using it to identify outbreaks or associate samples with them. That shaped the design more than any single number: the illness layer never touches an establishment, and Q9 records whose objection made that a rule.

Q3. The added datasets and their impact

We added four layers beyond the core, one for each remaining One Health domain and one to make a decision possible at all.

NOAA NCEI climate, GSOM monthly. Mean temperature, maximum temperature and precipitation for the 49 states and territories that hold FSIS-sampled poultry plants, averaged over a fixed panel of eight best-covered stations per state so the series is reproducible. The first version of the fetch used daily summaries and silently truncated at exactly 60,000 records per state; we caught it from the repeated round numbers in the output and rewrote it. The environment column of the system map.

FSIS recalls and public health alerts. Every poultry-or-Salmonella notice from the FSIS recall API, with the establishment mark recovered from the notice text where it is present. It is present for 206 of 563 notices, 36.6 percent, so a plant with no linked recall may simply be unresolvable. In the application, published recalls are pinned above anything the model produces, because a recall is a fact and a score is a forecast.

CDC/ATSDR Social Vulnerability Index 2022. County-level percentile ranks for all 3,144 US counties. This is what turns a food-safety tool into a One Health one: it is how the decision layer weighs who bears the consequence of a missed signal or a withheld pallet. The USDA food-access layer that requirement R6.1 asks for alongside it did not land, because the Census API key was rejected on every attempt, and we say so rather than let the requirement look satisfied.

A simulated lot inventory. A food bank's receiving log is not public data and we have none, so `14_build_synthetic_inventory.py` generates one from a seeded procedure: 1,880 lots over 52 weeks from 52 real plants, with weights, shelf lives and Arkansas destination counties drawn by stated rules. Every row carries `is_synthetic = True` and every figure derived from it carries a SIMULATED mark in the application. Without it there is no lot to decide about.

What they did to the risk model: nothing, and we measured it rather than assumed it. The leave-one-group-out ablation at six months, with establishment-clustered bootstrap intervals, finds only three FSIS feature groups that move AUROC beyond noise: the plant's own sampling history, the transparent projection, and all of the additions taken together. Serotype detail, the other pathogens FSIS tests, follow-up sampling, establishment profile, seasonality and product class are each within noise (`analysis/outputs/t19_ablation_h6.csv`). That tests FSIS's own columns. The brief asks about the data we added, so a second script adds the added layers to the full model on the same split and the same bootstrap (`05e_added_data_ablation.py`, `analysis/outputs/t19b_added_data_ablation_h6.csv`). Trailing three-month state climate, with a temperature anomaly against a climatology fit on training months only, does not help: AUROC 0.9191 against 0.9221, a change of minus 0.0030 with a 95 percent interval from minus 0.0065 to plus 0.0004. Linked recalls in the prior 24 months, resolved to 126 establishments through the mark on the notice, change nothing we can distinguish: plus 0.0009 with an interval from minus 0.0005 to plus 0.0026. Both together: minus 0.0032, interval minus 0.0067 to plus 0.0001. On a second build of the

same script the climate interval closed just below zero (minus 0.0065 to minus 0.0001); the point estimate is negative on both builds and we report it as within noise, leaning negative, rather than as a measured harm. At twelve months all three are within noise. At three months the recall layer is the one addition that shows a distinguishable gain on this build, plus 0.0033 with an interval from plus 0.0018 to plus 0.0051 (`t19b_added_data_ablation_h3.csv`); we do not quote three-month ablation verdicts as findings because they have flipped between builds (Q5), but it is the layer we would retest first, and it is already the one the alert queue ranks above every forecast. SVI is not tested there because it is a property of the destination county, not of the plant, and belongs to the decision layer.

So the plan to carry climate, burden, recalls and vulnerability as risk-model features was dropped, and the four layers went where they do change the answer: the decision layer. Recalls rank above forecasts in the alert queue. SVI drives the equity check and the food-security-weighted scenario. Illness burden weights which serotypes matter in the hazard view, at serotype level. The inventory is the thing being decided about. A longer feature table would have looked more impressive and said less; we tested whether the additions helped the model, they did not, and we used them where they help the decision.

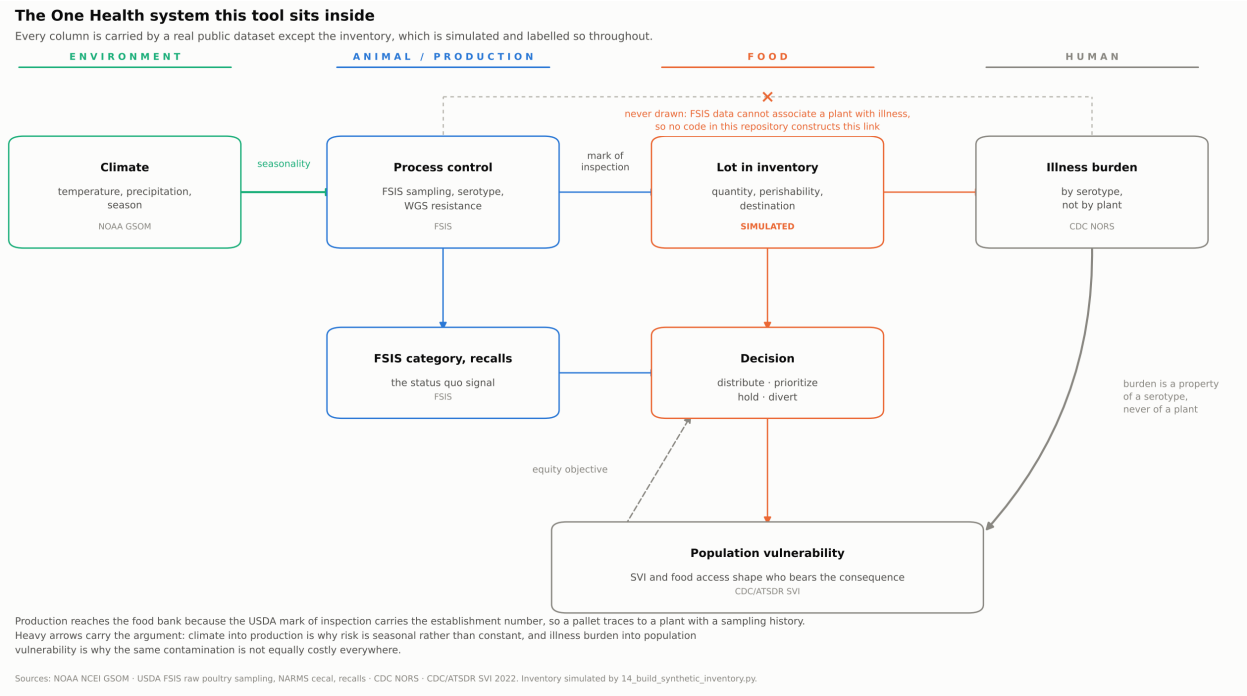


Figure 3: The One Health system map. Each dataset sits in its domain; the arrows that carry the argument are drawn; the establishment-to-illness arrow is drawn crossed out because the FSIS licence forbids it and no code in the repository constructs it. Built by 18_figure_system_map.py.

Q4. The modeled outcome

The unit is a plant and product class, one of the five FSIS Salmonella performance standards, observed on the 52-week moving window FSIS itself uses. The outcome is entry into FSIS

Category 3 within the forecast horizon: the point at which FSIS's own rating says a plant has exceeded its standard. We chose that outcome rather than a positivity rate because it is the status quo signal a food bank would otherwise act on, so the comparison to the baseline is on equal terms.

The category rebuild underneath this was validated before anything was modeled. Recomputing Category 1, 2, 3 and NA from the raw samples agrees with three official FSIS postings on 99.75 percent of 2,764 posted rows with zero false Category 3s (`analysis/outputs/category_validation.json`). The one rule FSIS does not document, how a borderline count rounds, was inferred empirically and is stated as such in the limitations.

Validation is time-aware. Training runs through September 2022; the test period begins in January 2024 and runs forward, so no window in the test set overlaps any window the models saw. At three months the test set holds 17,343 unit-months with 644 new Category 3 entries, a base rate of 3.71 percent (`analysis/outputs/early_warning_summary_h3.json`).

Three scorers were compared. The transparent baseline is a binomial projection: given the plant's current positive count and its sampling rate, the probability it crosses the Category 3 line within the horizon. Logistic regression and gradient boosting add sampling history, position against the standard, serotype detail, enumeration, the other pathogens FSIS tests, follow-up sampling, establishment profile, seasonality and product class. A leave-one-group-out ablation with establishment-clustered bootstrap intervals tested whether any of those additions carried signal (`analysis/outputs/t19_ablation_h6.csv`).

Which scorer the decision layer uses is chosen by measured calibration at the horizon it runs at, not by discrimination. A hold-or-divert decision acts on an absolute probability, so a score that ranks well but says 0.44 when the truth is 0.30 is the wrong score to act on. At three months the projection is far better calibrated than gradient boosting, expected calibration error 0.0056 against 0.0213, with the projection's top decile predicting 34.2 percent against 30.0 percent observed and gradient boosting's predicting 43.7 percent against 30.3 (`analysis/outputs/calibration_h3.json`). At six months the order reverses, 0.0092 against 0.0049. The decision layer reads the calibration file and picks, rather than carrying a hard-coded choice. Earlier in the project it did carry one, and the pipeline then ran it at the wrong horizon; that bug is why the choice is now data-driven.

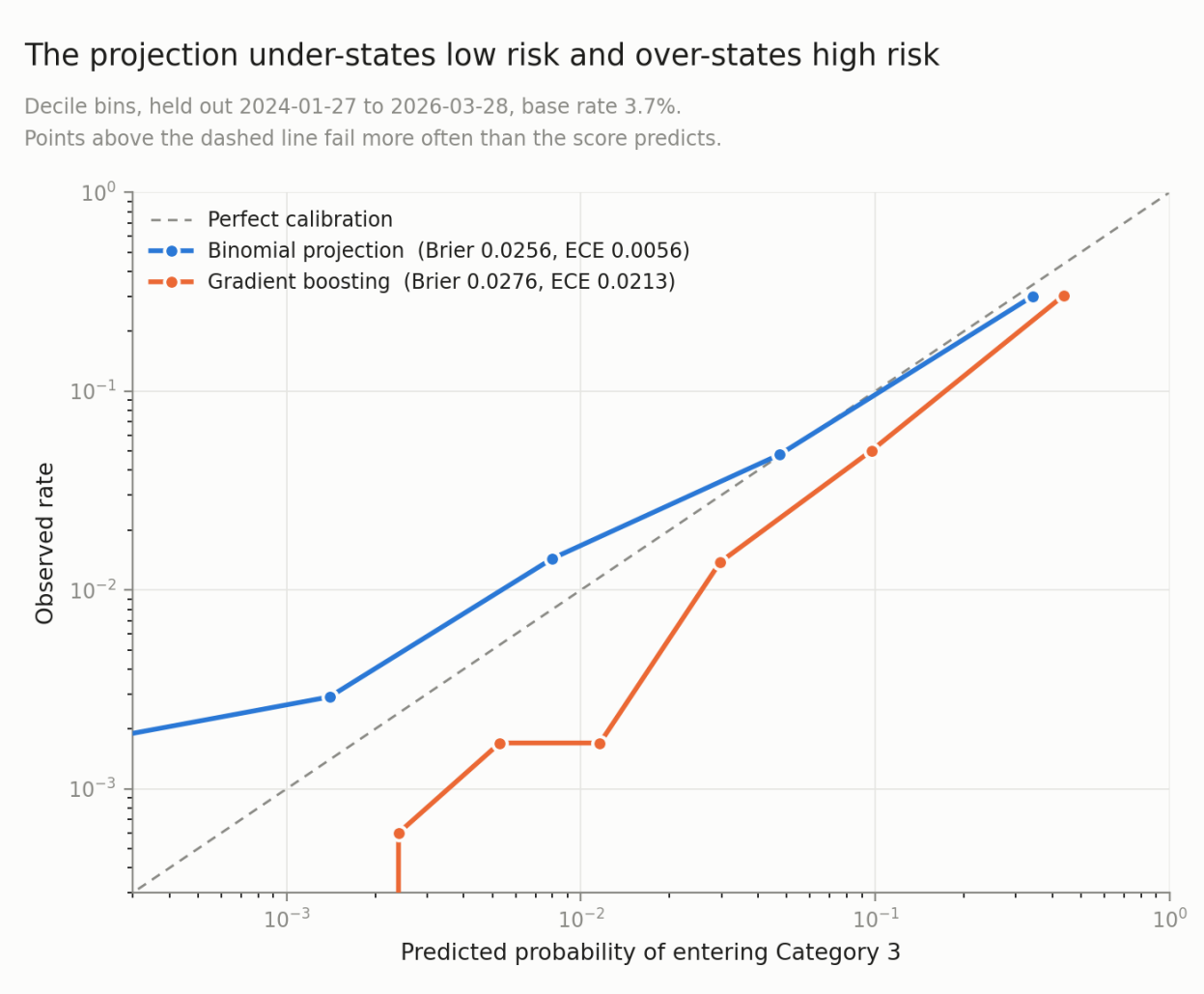


Figure 4: Reliability at the three-month horizon for both scorers, predicted against observed by decile, with ECE and Brier. The projection is the better-calibrated scorer at this horizon and is the one the decision layer uses. Built by 05c_calibration.py from t18.

Q5. Performance against the baseline

The required rule-based baseline is the one a food bank would use today: treat anything from a Category 2 or Category 3 plant as suspect. On the full panel at three months that flag reaches recall 0.915 and precision 0.145 while flagging 23.5 percent of all units (`analysis/outputs/early_warning_summary_h3.json`). It is a good detector and a poor allocator.

Against it, at six months, the projection scores AUROC 0.917 and gradient boosting 0.925. The difference is real but small: the establishment-clustered bootstrap over 578 plants resampled whole gives a median gain of 0.008 AUROC with a 95 percent interval from 0.000 to 0.016, and a median gain in average precision of 0.003 with an interval from minus 0.036 to plus 0.041, which is not distinguishable from zero (`analysis/outputs/early_warning_summary_h6.json`). We report that alongside the gain,

because the ablation explains it. Only three feature groups move AUROC beyond noise at six months: sampling history, the projection itself, and all of the additions taken together. Serotype detail, the other pathogens, follow-up sampling, establishment profile, seasonality and product class are each within noise (`t19_ablation_h6.csv`). The transparent baseline carries the signal.

Where the model earns its place is in allocation, not accuracy. A projection watch list the same size as the Category 2 list reaches recall 87.7 percent against the flag's 81.3 at equal precision; a list one tenth the size of the rated population reaches precision 41.5 percent at recall 66.0 (`analysis/outputs/watchlist_comparison_h6.json`). Lead time is usable: 86 percent of new Category 3 entries were on the top-10 percent list a month before they entered, and 62.6 percent three months before.

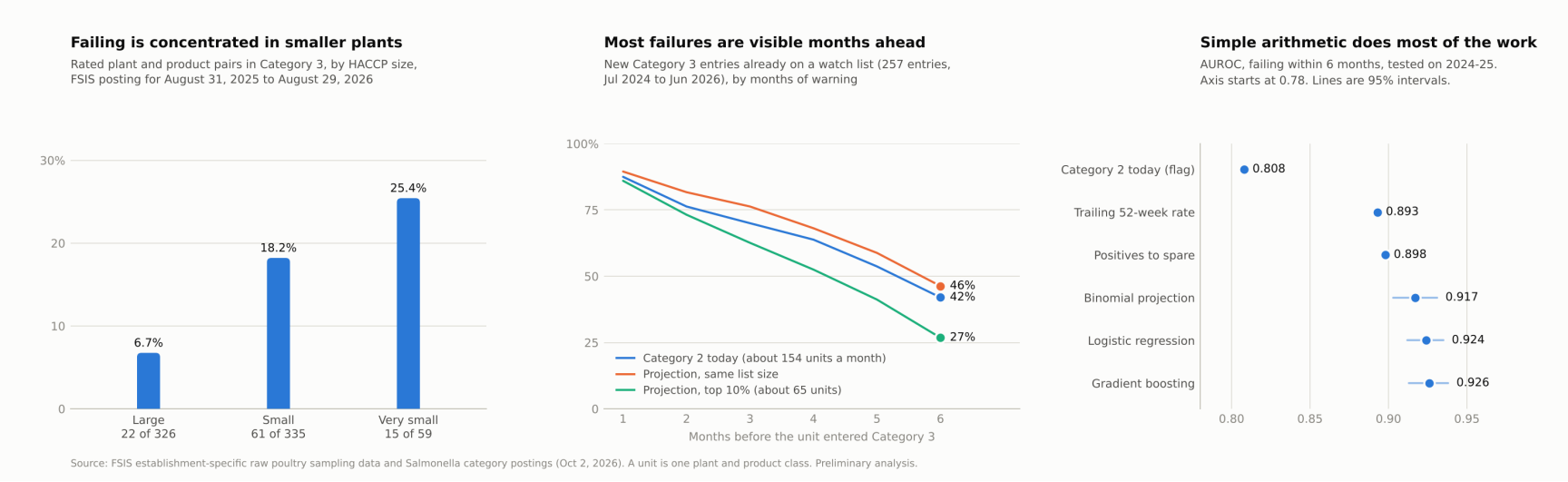


Figure 5: Who fails and how early. Share of rated pairs in Category 3 by plant size; how many months ahead new Category 3 entries were already on the top-10 percent watch list; and AUROC by scorer at six months with 95 percent establishment-clustered intervals. Built by 07_figures.py.

The three-month ablation verdicts are not reported because they are not stable. Two builds of the same data on two machines flipped `standard_position` between "carries signal" and "within noise", a consequence of 644 positives against 979 at six months and correspondingly wider intervals. The six-month ablation reproduces to the digit across machines, and the decision layer's outputs at three months reproduce to the digit as well. We quote what reproduces.

Q6. Which errors are costly: false alarms against missed risk

The two errors have to be in the same unit before they can be weighed against each other, so the decision layer reports both in meals at 1.2 pounds per meal. Avoidable loss is food destroyed from plants that did not go on to elevated risk: the cost of a false alarm, and a real loss. Exposure is food distributed uncooked from plants whose sampling later did deteriorate. Exposure is not a count of illnesses and must not be read as one; the FSIS data cannot attribute illness to a plant and we do not. It was called "missed risk" in an earlier version and renamed because that name implied a claim we cannot make.

On this supply base the status quo is the extreme case. Holding anything from a Category 2 or 3 plant catches every case, recall 1.000, and does so by holding 38 percent of incoming lots, which is 13.4 reviews a week against a modelled capacity of 6, and by destroying 108,499 meals of edible food, 91 percent of it from plants that never deteriorated (`analysis/outputs/t20_scenarios_h3.csv`). The conservative threshold policy catches 40 percent of the same risk and destroys 30,529 meals. That is the trade, and neither end of it is free.

Food is lost by two mechanisms, both mechanical and both derived from the simulated shelf life rather than from an assumed spoilage rate. A review hold consumes shelf life, and a lot that comes out of review with fewer days left than it takes to move through the pantry network is lost even though nobody discarded it. And diverted pounds beyond the cooked-meal channel's capacity are discarded. The first mechanism is what makes a conservative policy expensive, and Q8's second lot is an example of it.

The frontier between the two errors is not convex (`analysis/outputs/t22_frontier_h3.csv`). As the hold threshold rises from 0.02 to 0.10, avoidable loss falls from 61,095 meals to 8,861 while exposure barely moves, from 210,851 to 213,046. Then loss rises again, to 12,566 meals at 0.14, because raising the divert threshold pushes lots back into holds, where shelf life kills them. By 0.30 loss is down to 2,775 meals and exposure is up to 240,323. There is no point on that curve that is simply best. Choosing between 60,000 meals destroyed and 30,000 more meals of exposure is a judgement about whose risk counts, and the application shows the curve rather than picking a point on it.

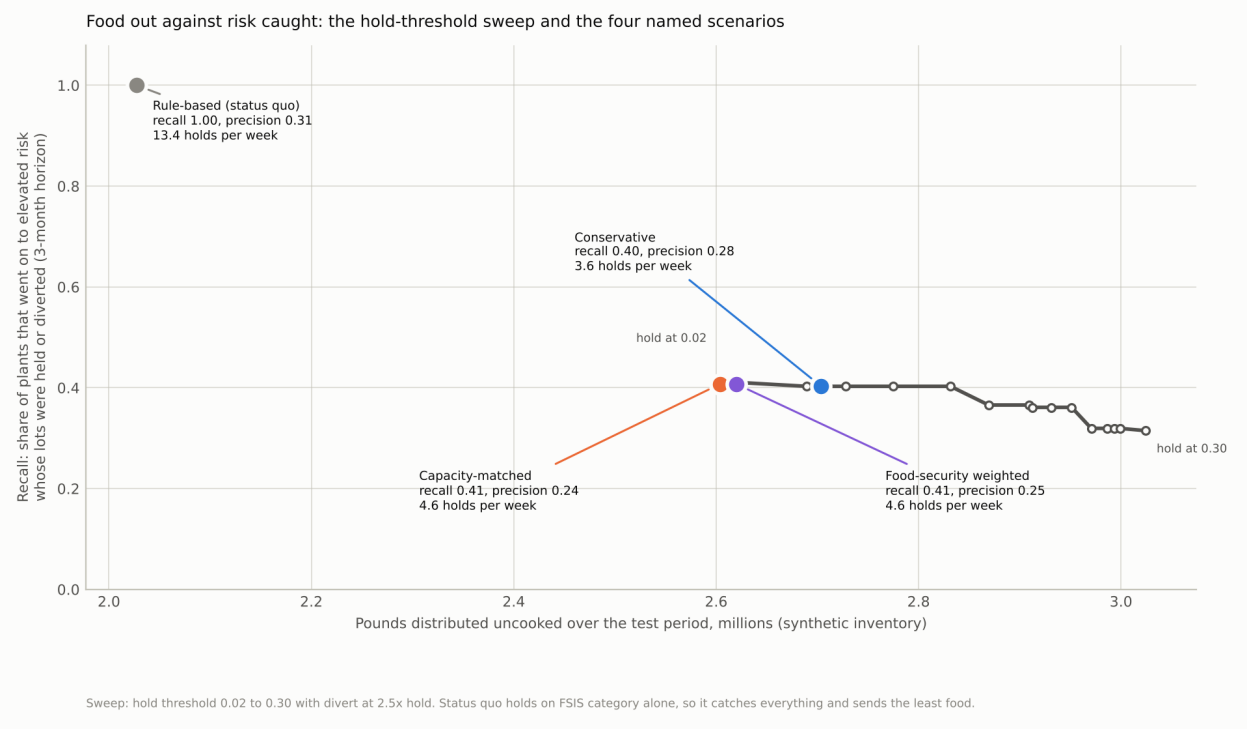


Figure 6: The frontier at three months: pounds distributed uncooked against recall as the hold threshold sweeps from 0.02 to 0.30, with the four named scenarios placed on it. Pounds are synthetic; recall and precision are measured on the FSIS test period. Built by 20_figure_decision_layer.py from t20 and t22.

One more result that surprised us. Setting the hold threshold to fill the review queue to capacity, which sounds like the sensible operational choice, loses to the simpler conservative threshold: 0.4 more points of recall for 54 percent more food destroyed, because every extra hold eats shelf life some lots do not have. Review capacity is not the binding constraint. Shelf life is.

Q7. Where food safety and food security conflict

Every withheld pallet has a destination, and the destinations are not equal. The decision layer joins each simulated lot's destination county to CDC/ATSDR SVI 2022 and reports, for every policy, how much food is withheld from the most vulnerable quartile of counties against the least, with the result stated whichever way it comes out (`analysis/outputs/t21_equity_by_svi_quartile_h3.csv`).

Under the conservative policy the withheld share is 14.7 percent of pounds in the most vulnerable quartile against 16.7 percent in the least, a gap of minus 1.96 points: holds fall slightly harder on low-need counties, which on this simulated supply base is luck rather than design. Under the rule-based status quo the gap is minus 0.28 points, no material difference. Under the capacity-matched policy it is minus 0.34.

The food-security-weighted scenario is the one that acts on the conflict directly. It requires a stronger signal before withholding from the most vulnerable quartile, raising the hold thresh-

old there by the weight and the divert threshold by half of it. The gap moves to minus 2.66 points, and the scenario destroys 4,971 fewer meals than the capacity-matched policy at identical recall, so on this data it dominates. The reason matters. It is close to free here because the counties being protected are not the ones supplied by the riskiest plants. Where those overlap, protecting supply to high-need counties means accepting more exposure in the places least able to absorb it, and the trade becomes real. The tool is built to show that gap on the scenario screen, not to optimise it away, and requirement R6.5 forbids it from doing so silently. The human chooses.

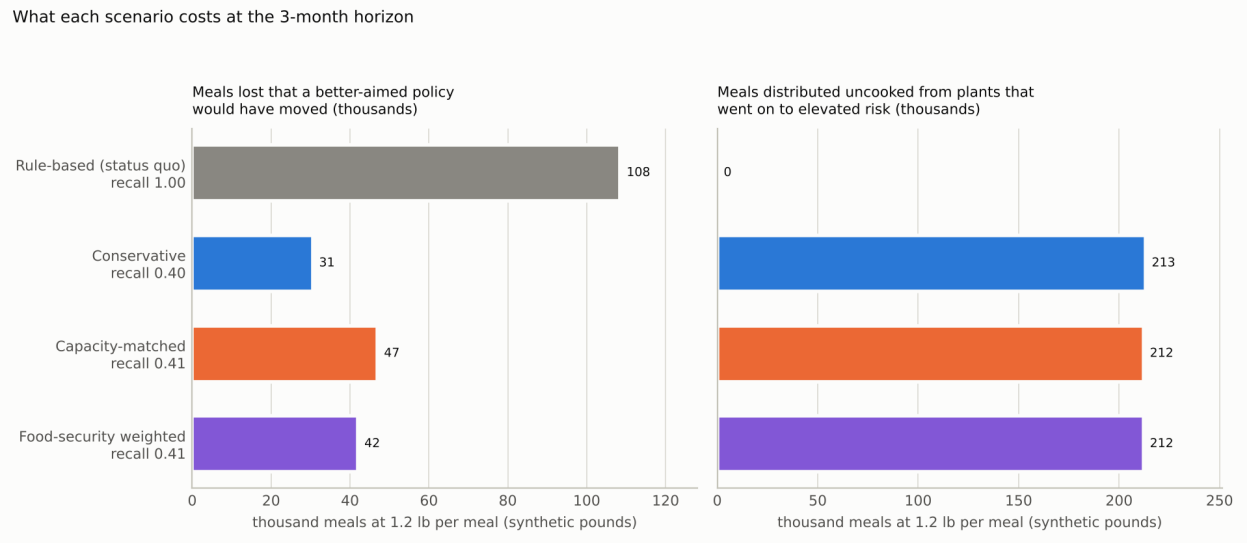


Figure 7: What each scenario costs at three months, in thousands of meals: food destroyed from plants that did not deteriorate (left) and food distributed uncooked from plants that did (right). The status quo has no exposure because it holds everything from a Category 2 or 3 plant, and destroys the most food doing it. Built by 20_figure_decision_layer.py from t20.

Two design choices made this conflict tractable rather than binary. Divert means the cooked-meal channel, so a withheld pallet is usually still food for someone; discard is the overflow, not the default. And the equity view reports the disparity in both directions, so a policy that happens to favour high-need counties is reported in the same terms as one that harms them. Both choices came from outside the implementation, and Q9 says from whom.

The limit is the one R6.1 points at: vulnerability here is county-level SVI alone, static and relative. The USDA food-access measure that would show where a withheld pallet has no substitute did not land. That is the first thing we would add.

Q8. An end-to-end scenario

Three lots from the receiving desk, taken as they appear in `analysis/outputs/t23_lot_actions_h3.csv` under the conservative scenario. The plants, their sampling histories, categories and scores are real. The lots themselves, their weights, dates and destination counties, are simulated, and carry that mark in the application.

The lot the tool was built for. On 5 March 2026 a pallet of comminuted turkey arrives,

4,608 pounds, bound for Pulaski County. The mark of inspection gives establishment 8371, Prestage Foods. The desk shows the call in large type: send it to the cooked-meal channel. The reason is one sentence: the plant's projection is at or above the divert threshold. Behind it, the evidence. FSIS currently rates the plant Category 2. In the last 52 weeks, 8 of 60 routine samples were positive. The projection puts the probability of the plant entering Category 3 within three months at 0.79, and the score is five days old. The calibration band the desk shows is the top decile of scores, which this lot is in: of plants scored in that decile, 30 percent went on to Category 3, against an average prediction of 34 percent for the decile, so the scorer is believable at that level. A score of 0.79 sits far out in the decile's tail, and the desk shows the band rather than pretending to a finer resolution than the data has. The lot has 124 days of shelf life, so there is no time pressure pushing the call either way. The desk states what the call costs: 4,608 pounds leave direct distribution to Pulaski County and go to a partner that cooks. It states the other choice: distribute now, which keeps the pounds in the pantry network and accepts the signal above. And it states what would reverse the call: the projection falling below 0.30, FSIS posting a Category 1 rating, or a clean 52-week window at the plant. Because divert is high-consequence, a dialog blocks until a person has read all of that and acknowledges it. Acknowledging changes nothing in the world. It records that a person read the call, and the next lot needing a decision comes up.

The lot that shows what a hold costs. On 3 February 2026 a pallet of chicken parts arrives from Case Farms, establishment 750, 2,621 pounds for Faulkner County. The plant is also Category 2, with 6 of 52 samples positive, but the projection is 0.08, far below the divert line and just above the conservative scenario's hold line of 0.05. The call is hold for review. The lot has nine days of shelf life. Review takes seven days, and getting a released lot through the pantry network takes three. Nine is less than ten. Under this policy the hold destroys the food, and the desk says so in the record: lost, shelf life exhausted by the review hold. Nobody discarded anything on purpose. This is the mechanism by which a conservative policy, which looks free when you count only what it catches, destroys 30,529 meals on this supply base, and it is what gives the frontier in Q6 its shape. A weak signal on a fresh lot is a decision between distributing it and losing it; holding it is not actually available.

The same plant, eight months earlier. On 12 July 2025 a smaller pallet of comminuted turkey arrived from the same Prestage plant, 1,454 pounds for Yell County. The most recent score for that plant and product was 350 days old, far past the 120-day staleness limit, so the desk treated the lot as unscored and distributed it with the gap named: no recent sampling for this plant and product. It did not treat an old score as a current one, and it did not treat the absence of a score as low risk. It said what it did not know. By March 2026 the same plant had 8 positives in 60 and a projection of 0.79, and the tool caught it. Whether it would have caught it earlier with fresher sampling is the question the limitations section addresses.

Those three lots are the project. A strong signal with time to act becomes a diversion that keeps the food in the system. A weak signal on a perishable lot is a choice the tool makes visible rather than a hold it pretends is free. And a missing score is reported as missing. Every number in them is in the application, and a reader can find each lot by its identifier on the receiving desk.

Q9. The effect of team science on the work

We are three people at two universities. Rushikesh Lagad (University of Arkansas, animal science with a focus on bioinformatics and genomics) planned the project and did the implementation: the FSIS pipeline, the model and its validation, the decision layer, the application, and the data and provenance work. Sambhavi Patel (University of Arkansas, microbiome bioinformatics) and Naisargi Modha (Northeastern University, microbiome bioinformatics) shaped what got built, and five of the decisions that define the project came from them rather than from the person writing the code. We record those here because the brief asks for it, and because without them this would be a different and worse tool.

The actor. The early framing aimed the tool at a regulator deciding where to send inspectors. That is the natural reading of the FSIS data if you come at it from the data side. Sambhavi proposed the regional food bank instead. Her argument was that the brief's three secondary objectives, food availability, food security and avoidable food loss, are side constraints for a processor or a retailer but are the actual mission of a food bank, and that a food bank is where the tension the brief asks about is lived every day. We rewrote the spec around that. Every requirement in `.kiro/specs/onehealth-decision-support/requirements.md` follows from it.

What divert means. The first version of the decision layer treated divert as a polite word for discard. Sambhavi pointed out that a plant with a bad sampling trajectory does not make its product inedible; it makes uncooked distribution the wrong channel for it, and food banks already have partners that cook on site. Divert became the cooked-meal channel, with a capacity, and discard became what happens when that channel is full. That one change is why the health objective and the loss objective can be traded against each other in the model instead of simply opposed, and it is what gives the frontier in Q6 a shape at all.

Never linking a plant to an illness. Joining the FSIS sampling data to CDC NORS outbreak data at the establishment level would have been easy and would have produced a striking table. Naisargi held that it was not acceptable: the FSIS data licence says the sampling data should not be used to associate samples with outbreaks, and the organizers' guardrails say the same in different words. Every join in the project is at serotype and year, the system map draws the forbidden arrow crossed out, and the app data build fails if an establishment identifier appears next to a burden figure. That constraint is now one of the four guardrails enforced in code, and it started as her objection.

Testing the added data instead of assuming about it. The first draft of this report argued that monthly climate would not help the model because month of year already did not, and left it there. Sambhavi read the brief's wording, an ablation showing the effect of the added data, and held that an inference from a neighbouring feature is not that. She wrote `05e_added_data_ablation.py` to add the climate and recall layers to the full model on the same split and bootstrap. The result agreed with the inference at six months, climate and recalls both within noise, but the paragraph in Q3 now reports a measurement with an interval rather than a prior, and it turned up the one thing the inference would have missed: a small, distinguishable gain from linked recalls at the three-month horizon on one build, which is the lead we would follow next. Her first version of the recall join linked two notices; review found the directory stores a plant's grants joined with a plus sign, and the corrected

join resolves 141 notices to 126 plants. Both versions are in the commit history.

The three-month horizon. The model was built and validated at three horizons, and the six-month horizon has the better headline numbers. Naisargi argued that the decision is made inside a lot's perishability window, which is days, and that the horizon should match the decision rather than the best-looking table. When we then checked coverage, 80 percent of lots had a usable score at three months against 64 percent at six, and the three-month policy was better on both recall and food loss. The numbers confirmed the reasoning, but the reasoning came first.

The pattern across all five is the same. The person closest to the data would have made the data-shaped choice. The people looking at the problem from the food bank's side and from the public-health side made the choices that turned a model into a decision tool. That division is not an accident of who did what; it is what the One Health framing is supposed to produce, and in this project it did.

These decisions are also logged, with dates, in the manual-change log and task 5.2 of `.kiro/specs/onehealth-decision-support/tasks.md`. We recorded them on 2026-10-29 from the team's own account rather than reconstructing them from the code afterwards.

Q10. What the tool can and can never claim, and what deployment would require

The tool can say which plants are on a deteriorating sampling trajectory, how far ahead that is visible, how well calibrated the number is, and what each policy for acting on it costs in shared units. Those claims rest on real public data and on results reproduced independently on two builds.

It can never say that any food, lot or plant is safe or unsafe. No output uses those words about anything, the application's test suite fails if one appears, and the only sentence in the whole system that contains them is the restriction that says we never use them. It can never link a plant to an illness. The FSIS data licence forbids using the sampling data to associate samples with outbreaks, and this project is built to respect that: every join between FSIS sampling and CDC NORS illness data is made at serotype and year, no code constructs an establishment-level link, the One Health system map draws that arrow crossed out with the reason on it, and the app data build fails if an establishment identifier appears beside a burden figure. It can never report a real poundage. A food bank's lot-level inventory is not public and we have none, so the inventory is generated by a seeded procedure, every row carries `is_synthetic = True`, and every pounds or meals figure in the application carries a SIMULATED mark that the build enforces. The ranking, the error rates and the direction of the equity effect do not depend on the simulation's scale; the absolute pounds depend on it entirely. And the quantity we call exposure, food distributed uncooked from plants whose sampling later deteriorated, is not a count of illnesses and must never be read as one. It was renamed from "missed risk" for exactly that reason.

Deployment would need five things this prototype does not have. Real inventory with lot-level provenance, under agreement with a food bank, which would turn every simulated

figure into a measurement. A cooked-meal partner with a stated capacity, so that diversion is a real channel and discard is a measured overflow rather than a modelled one. The USDA food-access layer that requirement R6.1 asks for alongside SVI, which did not land because the Census API key was rejected, so equity is currently judged on county-level SVI alone. A review workflow in which acknowledging a high-consequence recommendation does something, with the evidence that would reverse it attached, where today it records only that a person read it. And FSIS data fresher than the quarterly posting cycle; the median score a lot is matched to is 19 days old, and a lot whose plant has not been sampled within 120 days is treated as unscored rather than as low risk.

Everything above is a prototype in the organizers' sense: not a regulatory determination, not an outbreak declaration, and not a substitute for epidemiologic investigation.

Limitations

Each of these is documented at the place it arises; this collects them.

1. **The inventory is simulated.** Lots, weights, shelf lives and destinations come from a seeded generator, not a food bank. Every pounds and meals figure rests on it. The ranking, the error rates and the direction of the equity effect do not depend on the simulation's scale; the absolute pounds depend on it entirely.
2. **The decision layer's metrics describe a policy on 52 plants, not the model on 578.** The status quo's recall of 1.000 is a property of that narrow supply base. On the full panel at three months the Category 2 flag scores recall 0.915 and precision 0.145.
3. **Thresholds are horizon-specific and the scenario ranking is not horizon-robust.** A hold threshold is an absolute probability and means something different against a three-month outcome than a six-month one. At three months the capacity-matched policy is dominated by the food-security-weighted one; at six months the dominance runs the other way. Every such claim names its horizon.
4. **The three-month ablation verdicts are unstable between builds** and are not quoted. Six-month ablation and three-month decision-layer outputs reproduce to the digit on two machines.
5. **No establishment-level illness link, by licence and by design of the code.** Burden enters at serotype and year. NORS is outbreak-only and under-attributes; a serotype's burden can rest on a single event, and the application flags any figure where more than 80 percent of hospitalisations come from one outbreak.
6. **Food access is not in.** Vulnerability is county-level SVI 2022 alone: static, relative not absolute, and coarse for a service area that is part of a county. The Census key was rejected.
7. **The aminoglycoside gene heuristic overcalls.** It agrees with FSIS whole-genome resistance calls on 99.8 to 99.9 percent for beta-lactam, tetracycline, sulfonamide and phenicol, but only 62.3 percent for aminoglycoside. Every published resistance figure uses the FSIS calls, not the heuristic.
8. **Recall marks are recoverable for 36.6 percent of relevant notices.** A plant with no linked recall may be unresolvable rather than clear.
9. **The category rebuild infers one rule FSIS does not document**, how a borderline

count rounds. It matches 99.75 percent of posted rows, and the mismatches are listed per window in `analysis/outputs/category_mismatches_*.csv`.

10. **The establishment directory is a 2026 snapshot applied to earlier years.** Plants that changed hands or size are described as they are now.
11. **Enumeration is absent after 2026-04-28 and for turkey carcasses**, and the posting lag means the newest FSIS data is a quarter old. The median score a lot is matched to is 19 days old; a plant unsampled for 120 days is treated as unscored, not as low risk.
12. **The FSIS CDN refuses automated download** as of 2026-10-27. Reproduction is manual download plus an identical SHA-256 check, documented in the README.
13. **Predicting which plants fail is largely arithmetic.** The transparent projection carries the signal and the integrated model's advantage is not distinguishable from zero. We regard that as a finding, not a shortfall, but a reader expecting a modelling achievement should know it.

Team science statement

We are three graduate students at two universities. One of us works in animal science with a focus on bioinformatics and genomics; the other two work in microbiome bioinformatics, one at Arkansas and one at Northeastern, and came to the problem from the food bank's side and from the public-health side rather than from the FSIS data. One of us built the pipeline, the model, the decision layer and the application. The other two decided what it was for. The actor, the meaning of divert, the refusal to link a plant to an illness, the choice of the three-month horizon, and the insistence on measuring the added data's effect rather than inferring it all came from outside the implementation, and each is recorded in Q9 with the reasoning that was given at the time. The datasets we added changed the decision layer, not the risk model, and we report the two ablations that say so. The next thing we would add is a real receiving log from a regional food bank, under agreement, with lot-level provenance. It is the one addition that turns every simulated pounds and meals figure in this report into a measurement, and it would tell us whether the loss mechanisms we derived from shelf life and cooked-channel capacity are the ones that operate in practice. Roles, with dates, are in the manual-change log in `.kiro/specs/onehealth-decision-support/tasks.md`.

AI assistance disclosure

Kiro did not generate code in this repository. The `.kiro/` specs, steering files and hooks are written in Kiro's format and were loaded in Kiro, but every file in `.kiro/` was written by the team with Claude (Anthropic), and the manual-change log in `.kiro/specs/onehealth-decision-support/tasks.md` records what was changed by hand and when. Claude was used, under the team's direction, for the analysis pipeline (scripts 01 to 20), `17_build_app_data.py`, the React application and its tests, the data dictionary generator, the provenance file, and the first draft of the narrative in this report, which the team then edited. Every number in this report was computed by the pipeline, not by a language model, and every one names its source file. The team ran the pipeline on two machines, reviewed every output, and is responsible for all of it.